

AWS AI Job Search Agent

Profile-Isolated Job Discovery, Governance, ML Ranking & GenAI Review

AWS Portfolio Project - V2

Employer Adapters | Deterministic Governance | Lambda | SageMaker | Amazon Bedrock | S3 Website | Glue | MWAA | LangSmith

One shared AWS platform while search, ranking, publication and history remain isolated by candidate profile.

BUSINESS PROBLEM

- Employer sites expose different search backends and schemas.
- Candidate profiles require different job families and screening rules.
- Daily runs must avoid rescoring recent postings while still scoring every NEW eligible job.
- Recruiter-facing output needs traceable AWS / AI evidence.

ARCHITECTURE RESPONSE

- Shared platform + employer adapter layer.
- Central profile configuration; no title logic duplicated inside employer code.
- Deterministic screening before ML / GenAI.
- Profile-aware lifecycle and score cache.
- Daily publication with 7-day history.

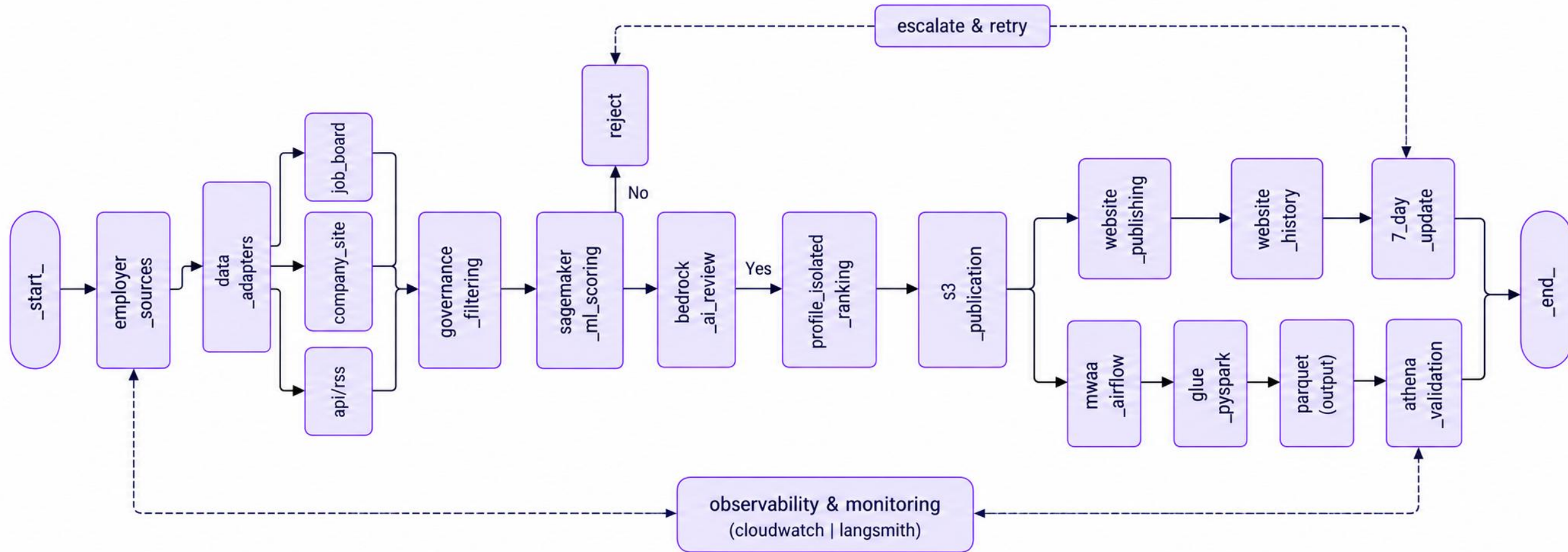
PROFILE ISOLATION

Frank C.
Data Science • AI • Quant • Analytics

W.Y. / Family-A
Bioanalysis • LC-MS/MS • Mass Spectrometry

Rule: profiles share infrastructure, not business search results.

Design principle: share infrastructure → isolate business intent → govern deterministically → score incrementally → publish auditable daily results.



Incremental processing reduces unnecessary inference while preserving profile isolation and deterministic controls.

1 DISCOVER

Employer adapters collect postings; profile configuration defines allowed job families.

2 GOVERN

Exclude internships, irrelevant roles and disallowed clearance requirements before AI scoring.

3 RESOLVE

Identify NEW eligible postings; reuse recent cached scores when allowed by lifecycle policy.

4 SCORE

Score every NEW eligible posting with SageMaker; retain model evidence for downstream selection.

5 REVIEW

Use Bedrock for grounded review while preserving deterministic governance boundaries.

6 PUBLISH

Rank high-to-low and retain up to Top 5 NEW jobs per employer; publish profile/date artifacts.

Daily contract: score all NEW eligible jobs first → then select Top 5 per employer → publish without cross-profile leakage.

Two profile-specific SageMaker XGBoost models provide quantitative job-fit scoring; Amazon Nova 2 Lite on Bedrock provides grounded semantic review.

1 Human / graded Ground Truth

2 Structured Features

3 XGBoost Training

4 SageMaker Inference

5 Job-Fit Score

6 Nova 2 Lite GenAI Review

7 Governed Ranking

FRANK C. — XGBOOST JOB-FIT MODEL

- Supervised XGBoost model trained in SageMaker.
- Ground truth: 24 labeled training rows; 8 human-approved positive rows.
- Structured fit features feed quantitative ranking.
- Real Batch Transform example: GDIT RQ227420 → score 0.8923.
- Interpretation guardrail: small-sample demonstration score, not a calibrated probability.

FAMILY-A / W.Y. — XGBOOST MODEL

- Separate profile-specific SageMaker XGBoost model for bioanalysis roles.
- Independent model + inference + prediction artifacts preserve profile isolation.
- Purpose: rank LC-MS/MS / bioanalysis opportunities using Family-A-specific relevance.
- Exact aggregate accuracy / calibration metrics are not claimed by the recovered evidence.

AMAZON BEDROCK — NOVA 2 LITE

- Amazon Nova 2 Lite is a transformer-based generative foundation model available through Amazon Bedrock.
- Project role: read JD + profile context and produce grounded semantic review.
- Outputs: verified matches, gaps / unknowns, concise assessment.
- Bedrock does not replace deterministic screening and does not overwrite the SageMaker score.
- Separation of duties: XGBoost ranks quantitatively; Nova explains semantically.

ML / GenAI design: human-review labels -> profile-specific XGBoost scoring -> grounded Nova 2 Lite review -> governed publication

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

jxc518 (288229085061)

Amazon SageMaker AI

Training jobs

job-search-frank-fit-20260928012513

Amazon SageMaker AI

Dashboard

What's new 41

Environment configuration

Applications and IDEs

Model training & customization

job-search-frank-fit-20260928012513

Clone

Create model package

Stop

Create model

Job settings

Job name

Status

SageMaker metrics time series

IAM role ARN

ARN

Creation time

Last modified time

Training time (seconds)

Billable time (seconds)

Managed spot training savings

Tuning job source/parent

Algorithm

Algorithm ARN

Additional volume size (GB)

Maximum wait time for managed spot trainings

Volume encryption key

Training image

Maximum runtime (seconds)

Managed spot training

Input mode

Instance group

Instance type

Instance count

Keep alive period

[Alt+S]

United States (N. Virginia)

jxc518 (288229085061)

Models

Create endpoint

Create endpoint configuration

Actions

Create model

Find models

1

Models table

Bedrock Marketplace (152)

Looking for latest open-weight and frontier models from Anthropic, OpenAI, and other providers? Click here to go to the new console optimized for bedrock-mantle and choose the region where the model is available.

Try the new Bedrock Console

Learn about bedrock-mantle endpoint

Spotlight

Find models

1 match

Recently released

Model name = Nova 2 Lite

Clear filters

Nova 2 Lite

By Amazon

Image, Video, Text to Text

Serverless

Cross-region inference

HR-ready website presents candidate-specific daily history while keeping the underlying platform shared.

FRANK C. — DATA SCIENCE / AI

September 29 daily publication

- Root summary: 3,154 raw jobs searched
- 191 passed deterministic governance
- 191 AI-scored NEW jobs
- 20 published Top jobs

Employers represented in the frozen daily artifact include Capital One, GDIT, Navy Federal and Optum.

W.Y. / FAMILY-A — BIOANALYSIS

September 29 daily publication

- Real employer/job records only
- AstraZeneca — Oncology Bioanalytical Strategy Lead
- Pfizer — Senior Scientist Mass Spectrometrists

September 28 was intentionally not backfilled because the only recovered Family-A artifact was a synthetic contract-test fixture.

Governance choice: missing historical evidence is preferable to publishing synthetic or unverifiable job data.

https://us-east-1.console.aws.amazon.com/gluestudio/home?region=us-east-1#/editor/job/job-search-published-jobs-analytics/graph

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) jxc518 (288229085061) jxc518

job-search-published-jobs-analytics Last modified on 10/3/2026, 3:51:21 PM Actions Save Run

Visual Script Job details Runs Data quality Schedules Version Control

+ Data source - S3 bucket Amazon S3 Producti... Transform - Dynamic Tra... Flatten Published Jobs Transform - SQL Query Extract Published Jobs Transform - Change Sch... Select Analytics Fields Transform - Change Sch... Select Published Job ... Data target - S3 bucket AWS S3 Published

Transform

Name: Select Published Job Fields

Node parents: Choose one or more parent node

Extract Published Jobs X
SqlCode - Transform

Change Schema (Apply mapping)

Source key	Target key	Data type	Drop
snapshot_date	snapshot_	string	☐
profile_id	profile_id	string	☐
job_id	job_id	string	☐
company_id	company_	string	☐
company_name	company_	string	☐
title	title	string	☐

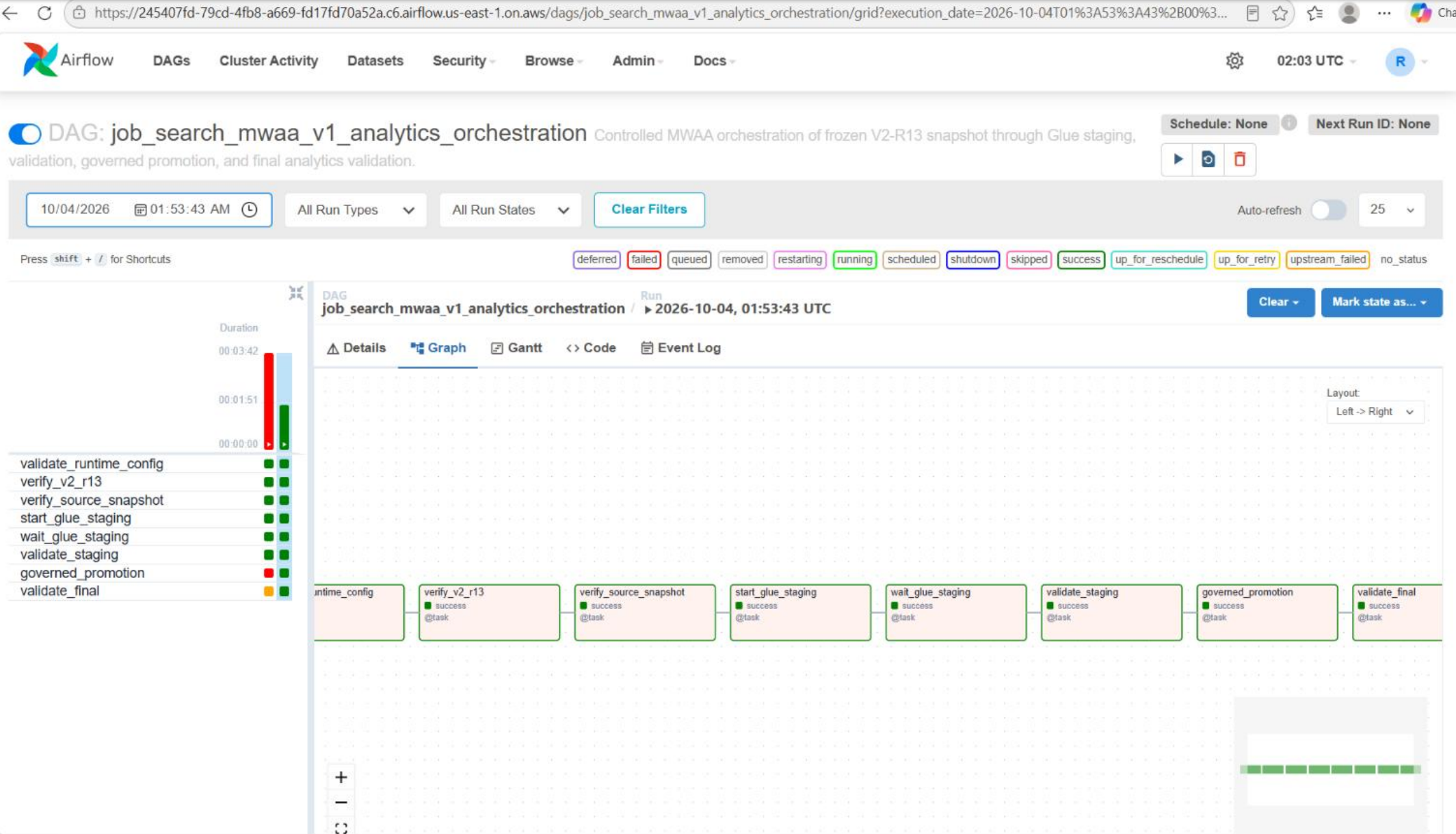
Data preview (30) Info READY End session Previewing 25 of 25 fields

Filter sample dataset

snapshot_date profile_id job_id company_id company_name

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

S3 daily snapshots → schema transformation → partitioned Parquet → Glue Data Catalog → Athena validation



Dependency Management | Retry & Failure Handling | Run-Scoped Staging | Governed Promotion | Final Validation

https://us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1



CloudWatch > Log management > airflow-job-search-mwaa-v1-Task

CloudWatch New

Favorites and recents

Ingestion

Dashboards

Alarms 1 2 0

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management New

Log Analytics New

Log Anomalies

Log events

You can use the filter bar below to search

Display

Timestamp

2026-10-04T01:55:30.627Z

[2026-10-04T01:55:30.627+0000 bytes=12066]

https://245407fd-79cd-4fb8-a669-fd17fd70a52a.c6.a



DAGs Cluster Activity Datasets

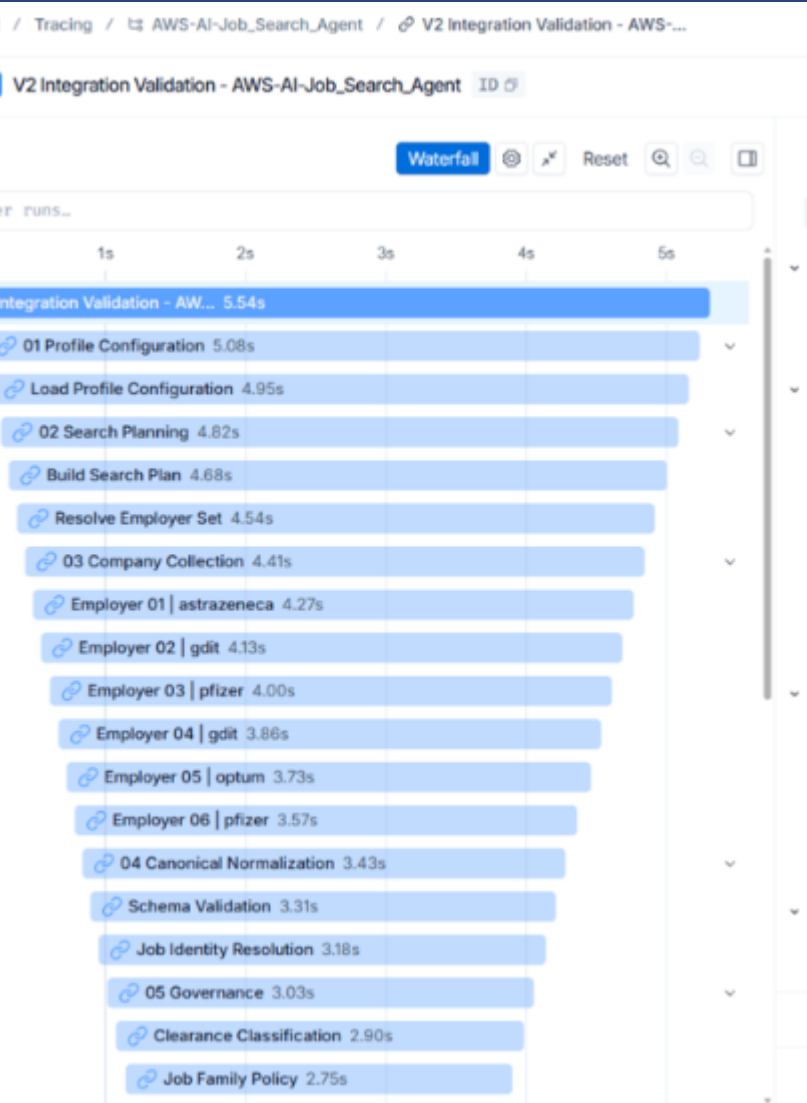
DAG: job_search_mwaa_v1_analy

validation, governed promotion, and final analytics validation

All Run Types

Press **shift** + **/** for Shortcuts

	Duration	
	00:03:42	
	00:01:51	
	00:00:00	
validate_runtime_config		
verify_v2_r13		
verify_source_snapshot		
start_glue_staging		
wait_glue_staging		
validate_staging		
governed_promotion		
validate_final		



AWS Runtime — CloudWatch · MWAA_E2E_PASS

Workflow — Airflow · 8/8 Tasks Passed

AI / Agent — LangSmith · Multi-Stage Trace